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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

Step 1:

This step is purely descriptive and does not involve code execution. I will reiterate the definitions as provided previously.

Market Risk Premium (Mkt-Rf)

Definition: The Market Risk Premium (Mkt-Rf) is the excess return of the value-weighted market portfolio over the risk-free rate. It represents the additional return investors expect for taking on the systematic risk of the overall stock market.

Why this factor helps to explain returns (Non-technical paragraph): Investors generally demand a higher return for investments that carry more risk. The market risk premium captures this fundamental concept: if you invest in the broad stock market, which is inherently volatile, you expect to be compensated beyond what you would earn from a risk-free asset like a Treasury bill. This extra return encourages investors to allocate capital to productive enterprises, driving economic growth. It reflects the overall compensation for bearing non-diversifiable market-wide risks that cannot be eliminated by simply holding a diversified portfolio.

Professional References :

- Fama, E. F., & French, K. R. (1992). The Cross-Section of Expected Stock Returns. *The Journal of Finance*, 47(2), 427-465.
- Grinblatt, M., & Titman, S. (2000). *Financial Markets and Corporate Strategy*. McGraw-Hill/Irwin. (General finance textbook reference for CAPM and market risk premium).

Small Minus Big (SMB)

Definition: Small Minus Big (SMB) is a size factor, calculated as the average return on portfolios of small-cap stocks minus the average return on portfolios of large-cap stocks. It captures the historical tendency for small-capitalization companies to outperform large-capitalization companies.

Why this factor helps to explain returns (Non-technical paragraph): Historically, smaller companies have often delivered higher returns than larger, more established companies. This could be because small firms are perceived as riskier or less liquid, and investors demand extra compensation for holding them. Alternatively, small companies might have greater growth potential or be undervalued by the market due to less analyst coverage. This factor suggests that investing in smaller companies, on average, can provide a premium return.

Professional References :

- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56.
- Chan, L. K., Hamao, Y., & Lakonishok, J. (1991). Fundamentals and stock returns in Japan. *The Journal of Finance*, 46(5), 1739-1764. (Supports size effect in other markets).

High Minus Low (HML)

Definition: High Minus Low (HML) is a value factor, calculated as the average return on portfolios of high book-to-market equity (value) stocks minus the average return on portfolios of low book-to-market equity (growth) stocks. It reflects the tendency for "value" stocks (those with high book-to-market ratios, often considered undervalued) to outperform "growth" stocks (those with low book-to-market ratios).

Why this factor helps to explain returns (Non-technical paragraph): Value stocks, typically mature companies with stable earnings but perhaps limited growth prospects, have historically outperformed growth stocks, which are often younger companies with high potential but also higher risk. This could be because value stocks are perceived as less risky or are simply "cheaper" relative to their fundamentals. Investors are compensated for buying these overlooked or out-of-favor companies.

Professional References:

- Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56.
- Lakonishok, J., Shleifer, A., & Vishny, R. W. (1994). Contrarian Investment, Extrapolation, and Risk. *The Journal of Finance*, 49(5), 1541-1578. (Discusses the value premium).

Conservative Minus Aggressive (CMA)

Definition: Conservative Minus Aggressive (CMA) is an investment factor, calculated as the average return on portfolios of companies that invest conservatively (low asset growth) minus the average return on portfolios of companies that invest aggressively (high asset growth). It posits that companies with conservative investment policies tend to outperform those with aggressive investment policies.

Why this factor helps to explain returns (Non-technical paragraph): Companies that invest more cautiously, perhaps by retaining earnings or growing assets at a slower pace, tend to be more profitable and financially sound in the long run. Conversely, companies that invest aggressively, often through rapid expansion or large capital expenditures, might be prone to overinvestment or less efficient capital allocation. Investors are rewarded for favoring companies that exhibit disciplined and conservative investment strategies.

Professional References :

- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1-22.
- Titman, S., Wei, K. J. and Xie, F. (2004). Capital Investments and Stock Returns. *Journal of Financial and Quantitative Analysis*, 39(4), 677-700. (Related research on investment and returns).

Robust Minus Weak (RMW)

Definition: Robust Minus Weak (RMW) is a profitability factor, calculated as the average return on portfolios of companies with robust profitability (high operating profitability) minus the average return on portfolios of companies with weak profitability (low operating profitability). It suggests that more profitable companies tend to generate higher returns for investors.

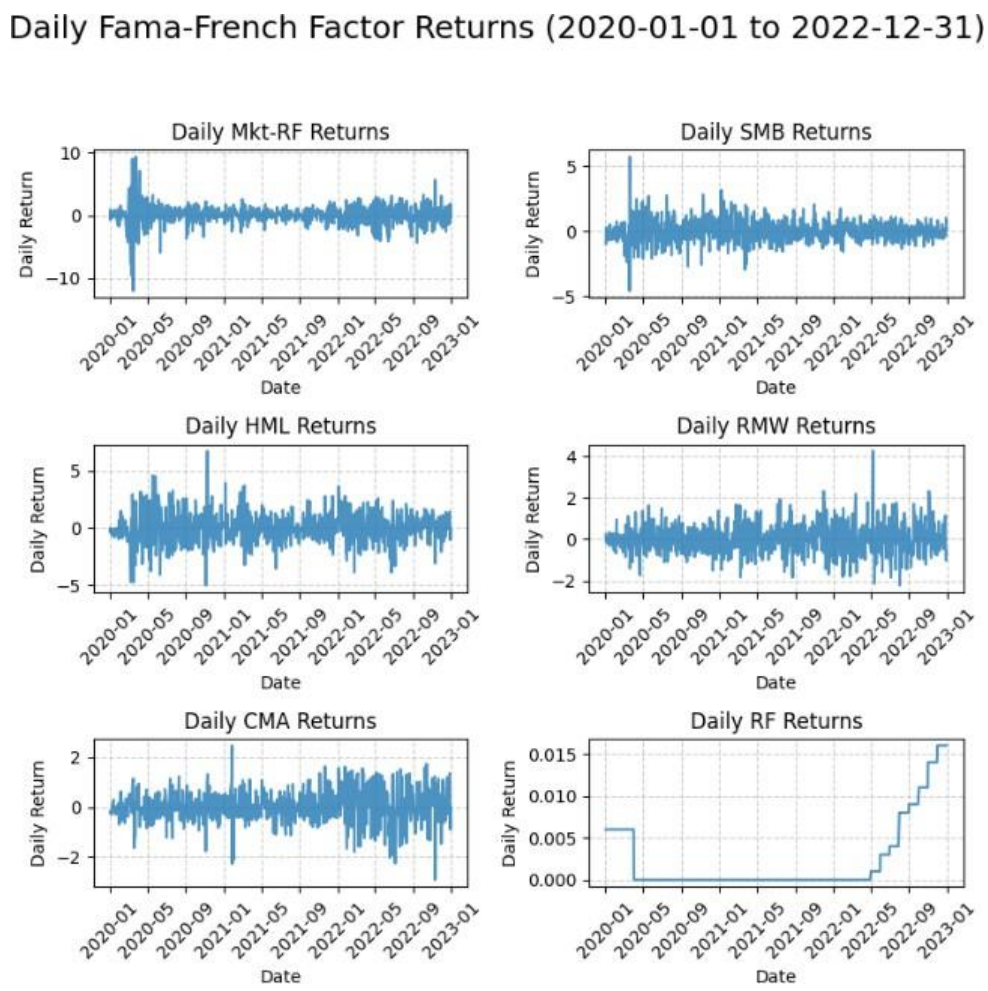
Why this factor helps to explain returns (Non-technical paragraph): Companies that are consistently more profitable, evidenced by strong operating margins, tend to be better investments. These companies often have sustainable competitive advantages, efficient operations, and a greater ability to generate cash flow. Investors are compensated for investing in these financially healthy and robust businesses, as their consistent earnings power contributes to superior stock performance over time.

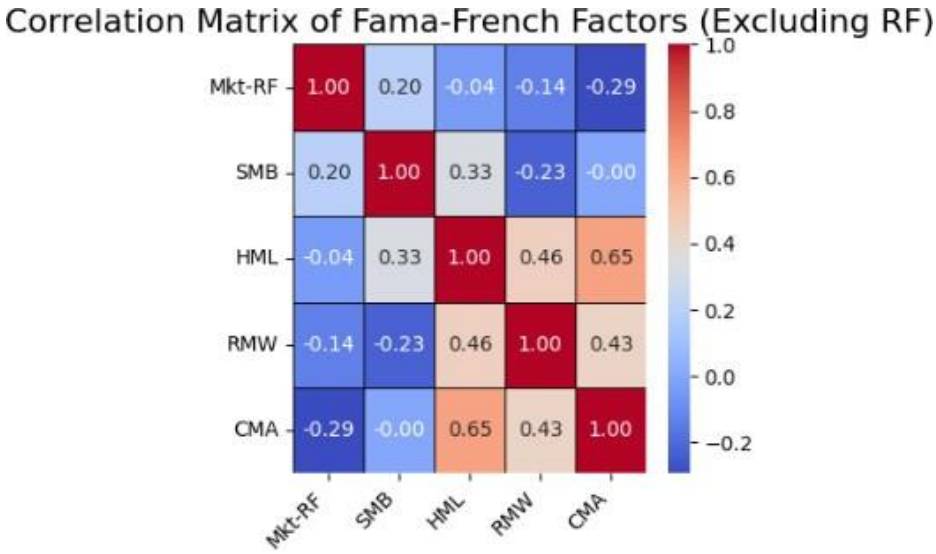
Professional References :

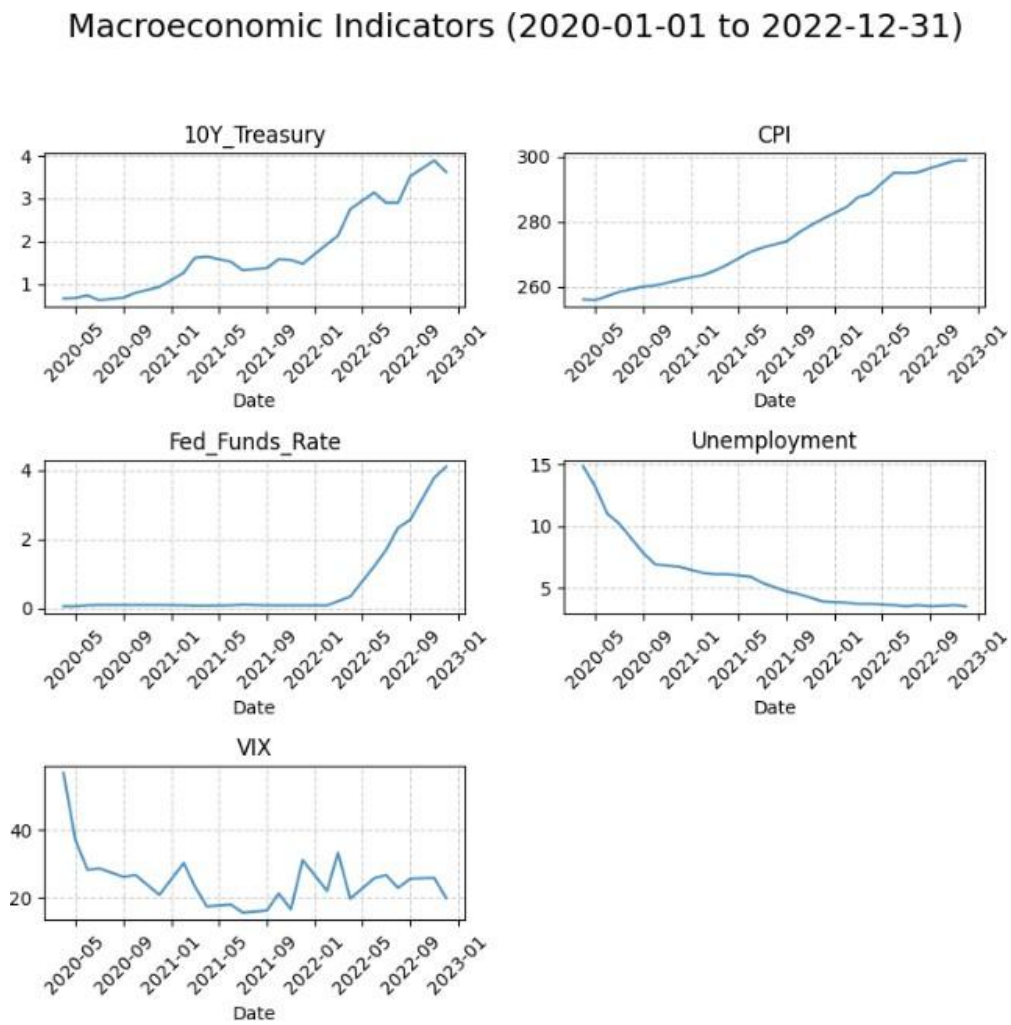
- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1-22.
- Novy-Marx, R. (2013). The Other Side of Value: The Gross Profitability Premium. *Journal of Financial Economics*, 108(1), 1-20. (Provides a basis for profitability as a factor).

Step 2:

This step focuses on understanding the Fama-French factor data within the specified 3-year period (2020-2022). We begin by visualizing the daily returns of each factor (Market-Rf, SMB, HML, RMW, CMA, RF) to observe their time-series behavior. Following this, a correlation matrix is computed and visualized using a heatmap. The correlation matrix is crucial for understanding the relationships between the factors: low correlations suggest that factors capture distinct sources of risk and return, while high correlations might indicate redundancy or shared underlying drivers.




Correlation Matrix of Fama-French Factors (Excluding RF)
Mkt-RF
SMB
HML
RMW
CMA
1.00 0.20 -0.04 -0.14 -0.29
0.20 1.00 0.33 -0.23 -0.00
-0.04 0.33 1.00 0.46 0.65
-0.14 -0.23 0.46 1.00 0.43
-0.29 -0.00 0.65 0.43 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Macroeconomic Indicators (2020-01-01 to 2022-12-31)
10Y_Treasury
CPI
Fed_Funds_Rate
Unemployment
VIX
Date
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

Step 3:

Step 3 involves implementing the Fama-French 3-Factor model. First, we download historical adjusted close prices for a sample stock (Apple Inc. - AAPL in this case) and calculate its daily returns. These returns are then converted to *excess returns* by subtracting the risk-free rate provided in the Fama-French data. The data is then split into training (80%) and testing (20%) sets based on time series.

Two types of regressions are performed on the training data: Ordinary Least Squares (OLS) and Robust Regression (using Huber's T-estimator). OLS is standard, while Robust Regression is used to mitigate the influence of outliers in the data. Finally, the models are evaluated on both training and testing data using Mean Squared Error (MSE) and Mean Absolute Error (MAE) to assess their predictive performance.

```
--- FF3: Model Evaluation on Training and Testing Data ---
FF3 OLS Train MSE: 0.00014853, MAE: 0.00876883
FF3 OLS Test MSE: 0.00022760, MAE: 0.01227230
FF3 Robust Train MSE: 0.00014889, MAE: 0.00874398
FF3 Robust Test MSE: 0.00022342, MAE: 0.01211361
FF3 OLS Train R-squared: 0.73181885
FF3 OLS Test R-squared: 0.59195899
```

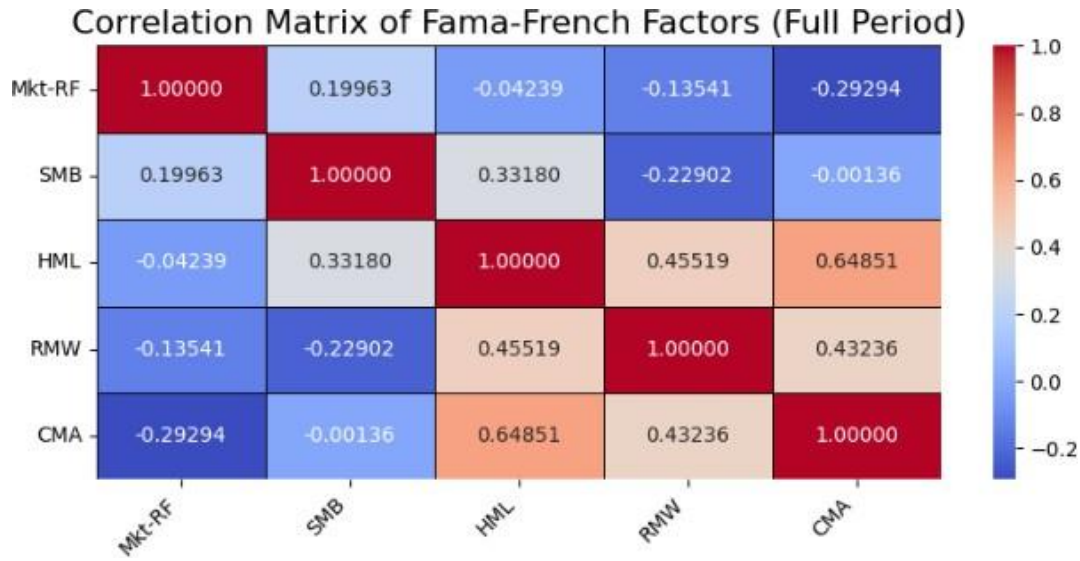
Step 4:

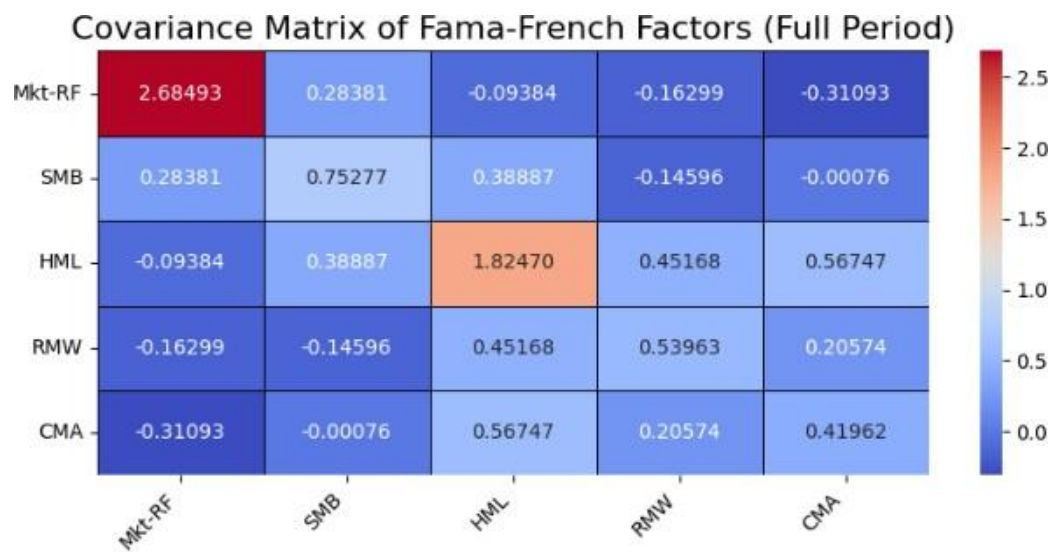
Building upon the previous step, Step 4 implements the Fama-French 5-Factor model. This model extends the FF3 by adding two new factors: Robust Minus Weak (RMW) for profitability and Conservative Minus Aggressive (CMA) for investment. The same sample stock (AAPL) and the same train-test split are used for consistency. We again perform both OLS and Robust (Huber) regressions on the training data.

The addition of RMW and CMA allows for a more comprehensive assessment of how these factors influence the stock's excess returns. Model evaluation, including MSE and MAE for both training and testing sets, is conducted to compare its performance against the FF3 model.

```
--- FF5: Model Evaluation on Training and Testing Data ---
FF5 OLS Train MSE: 0.00010758, MAE: 0.00743905
FF5 OLS Test MSE: 0.00026921, MAE: 0.01319162
FF5 Robust Train MSE: 0.00010790, MAE: 0.00741116
FF5 Robust Test MSE: 0.00026087, MAE: 0.01299721
FF5 OLS Train R-squared: 0.80575983
FF5 OLS Test R-squared: 0.51736012
```

Step 5:





While both covariance and correlation are used to measure the co-movement between variables, they are different in many ways:

1. Difference of range
   - Correlation is scaled and the coefficient ranges from -1 to +1. A positive value indicates same direction movement and a negative value suggests opposite directions, Furthermore, correlation of 0 means no linear relationship between the two variables.
   - The covariance matrix is an unscaled measure of how two factors move together. A positive covariance means that the two variables tend to move in the same direction, and a negative value means opposite directions.

- It can be seen that both matrices always show the same sign for the corresponding pair. This is because correlation is a scaled version of covariance. However, the values are different and are not directly comparable without scaling.

2. Difference of units in making comparison
   - Correction is scaled and therefore easier to compare the strength of relationship between pairs.
   - Covariance is unscaled, the unit of coefficient is dependent on the underlying data. Therefore, it is more difficult to compare the strength or magnitude.
   - This can be evident when looking at the diagonal element comparison. In the correlation matrix , the value for Mkt-RF and Mkt-Rf is 1, indicating a perfect positive relation. This is expected because they are identical variables and should have perfect correlation. Whereas the covariance for the same pair is 2.68, and shows the variance of the factors and is an indicator for volatility and not relationship strength.
3. Difference in interpretability
   - It is therefore easier to interpret correlation matrix than the covariance matrix as correlation coefficient is standardized with a fixed range and removes the impact of units.
   - While correlation is very good for comparing the relationship strength, covariance is directly used in portfolio optimization because the scale of the variable matters in the calculation. It is also used in the risk assessment because it is a measure of the volatility for each factor.

In our model, we can see that:

1. HML and CMA have the highest correlation coefficient (0.64851), showing a strong positive correlation.
2. CMA and SMB have the lowest correlation coefficient of (-0.00136), suggesting that the two factors are uncorrelated and are independent.
3. Mkt-RF is the most volatile factor that showed a variance of 2.68493, followed by HML of 1.82470. The higher variance suggests that the two factors may have contributed more to the model and could explain a larger proportion of variance in prediction than other factors.

Step 6:

```
--- Step 6: Tabular Comparison of Model Results ---
Model                        FF3 OLS  FF3 Robust   FF5 OLS  FF5 Robust
Alpha (const)               0.000208   -0.000063 -0.000416   -0.000478
Mkt-RF                      0.012095    0.012071  0.013087    0.013178
SMB                        -0.004495   -0.004066 -0.001657   -0.001401
HML                        -0.002656   -0.002482 -0.007870   -0.007608
RMW                              NaN         NaN  0.004404    0.004322
CMA                              NaN         NaN  0.013177    0.012442
Alpha p-value               0.676252    0.884380  0.329691    0.192658
Mkt-RF p-value              0.000000    0.000000  0.000000    0.000000
SMB p-value                 0.000000    0.000000  0.003786    0.004289
HML p-value                 0.000000    0.000000  0.000000    0.000000
RMW p-value                      NaN         NaN  0.000000    0.000000
CMA p-value                      NaN         NaN  0.000000    0.000000
Train R-squared (or Pseudo) 0.731819         NaN  0.805760         NaN
Test R-squared (or Pseudo)  0.591959         NaN  0.517360         NaN
Adjusted R-squared          0.730478         NaN  0.804136         NaN
Train MSE                   0.000149    0.000149  0.000108    0.000108
Test MSE                    0.000228    0.000223  0.000269    0.000261
Train MAE                   0.008769    0.008744  0.007439    0.007411
Test MAE                    0.012272    0.012114  0.013192    0.012997
```

FF3 models consist of three factors: Mkt-RF, SMB and HML, and the FF5 models include all the factors from the FF3 model and additional two factors: CMA and RMW.

With the two additional factors, FF5 showed a better performance in the training dataset. Looking at the OLS model, including the two additional factors increased the R-squared from 0.732(FF3) to 0.806(FF5). The slightly higher R-squared from FF5 means it was able to explain around 7.4% more variance in the stock return. Furthermore, the training error (MSE) was reduced at the same time from 0.000149(FF3) to 0.000108 (FF5).

However, FF5 performed worse than FF3 in the test dataset, with the R-squared decreased from 0.592(FF3 OLS) to 0.517 (FF5 OLS) and the Mean Square Error (MSE) increased from 0.000228 (FF3 OLS) to 0.000269(FF5 OLS). This means the FF3 OLS model was able to make more accurate predictions on unseen data and has a higher generalisability than FF5 OLS model. This is also an indicator for overfitting in FF5. As the FF5 incorporates more factors, the model becomes more complex and is exposed to more noise that is specific to the training dataset.

This result is not surprising and aligns with our observation from Step 5e: Mkt-RF and HML are the most contributing factors to the explained variance in the models. As FF3 has already included the most influential factors Mkt-RF and HML, any additional factor will have diminished and limited contributions.

However, this does not mean the two additional factors, CMA and RMW, had no value in FF5. In the FF5 OLS model, CMA and RMW have coefficients 0.0044 and 0.0132 respectively, and both were highly significant (p-value <0.0000001). It is important to point out that their coefficient was higher than the SMB

(-0.0017, p-value = 0.003786). SMB's coefficient was reduced significantly in FF5 model compared to that in the FF3 model (-0.0045, p-value <0.0000001). This suggests that CMA and RMW may capture similar or overlapping information, reducing SMB's unique contribution to the model.

In conclusion, the two additional factors, CMA and RMW, in the FF5 model:

1. improves the explained variance in the training dataset at the cost of generalisability to unseen dataset.
2. Were more powerful factors than the SMB in the model, and they diminished SMB's unique contribution to the model.
3. Future work can investigate the impact of removing SMB as a way to improve prediction performance and generalisability.

Step 7: Portfolio Optimization with Fama-French Factors

Part 1: Data Acquisition

Data Acquisition (yfinance):

- Using the yfinance library to download historical "Close" prices for your chosen tickers.
- Calculating daily percentage returns for each stock.
- Calculates stock_excess_returns by subtracting the daily risk-free rate (RF) from the stock returns.

```
/tmp/ipython-input-2-1378305196.py:12: FutureWarning: YF.download() has changed
  stock_data = yf.download(tickers, start=start_date, end=end_date)['Close']
[*********************100%***********************]  5 of 5 completedStock Returr
 Ticker          AAPL       JNJ       JPM      MSFT       XOM
Date
2021-07-02  0.019596  0.018197 -0.000961  0.022275 -0.001423
2021-07-06  0.014719 -0.005977 -0.016792  0.000037 -0.028494
2021-07-07  0.017955  0.008573  0.001173  0.008175 -0.015643
2021-07-08 -0.009200 -0.001948 -0.017254 -0.008966 -0.004470
2021-07-09  0.013055  0.003962  0.031999  0.001874  0.018125
Stock Returns Shape: (752, 5)
FF Factors Head:
              Mkt-Rf       SMB       HML       RMW       CMA            RF
Date
2021-07-02  0.004381 -0.003731 -0.005908  0.001887 -0.001304  1.576081e-05
2021-07-06 -0.005170  0.001411  0.000341 -0.002220  0.000537  5.297124e-07
2021-07-07 -0.015537  0.006430  0.005496 -0.000439 -0.002501  1.773644e-05
2021-07-08  0.004058 -0.007426  0.007458  0.000088  0.001112 -1.941912e-06
2021-07-09  0.002809 -0.005162  0.003462  0.005296  0.003903  1.265152e-05
FF Factors Shape: (752, 6)
```

2. **Markowitz Portfolio Optimization (scipy.optimize.minimize):**

- Expected Returns (mu): Calculates the historical average daily excess return for each stock.
- Covariance Matrix (sigma): Computes the pairwise covariance of daily excess returns among your 5 stocks. This is the core input for risk calculation in Markowitz.
- Objective Functions:
    - portfolio_variance: Calculates the variance of a portfolio given weights and the covariance matrix.
    - portfolio_return: Calculates the expected return of a portfolio given weights and expected returns.
    - neg_sharpe_ratio: This is the function we *minimize* to *maximize* the Sharpe Ratio. It takes the portfolio return, standard deviation, and a risk-free rate.
- Constraints: {'type': 'eq', 'fun': lambda x: np.sum(x) - 1} ensures that the sum of the portfolio weights equals 1 (meaning 100% of capital is allocated).
- Bounds: bounds = tuple((-1, 1) for _ in range(num_assets)) allows for short selling. A weight of -0.20 for a stock means you are short 20% of the portfolio value in that stock. If you wanted no shorting, you'd use (0, 1) for each bound.
- minimize function: This is the core optimization solver. It finds the set of weights that minimizes the neg_sharpe_ratio (or portfolio_variance for min-variance portfolio) subject to the defined constraints and bounds.
- Output: Prints the optimal weights for the Max Sharpe Ratio portfolio and the Minimum Variance portfolio, along with their associated returns and standard deviations.

```
--- Optimal Portfolio (Max Sharpe Ratio) ---
Optimal Weights:
  AAPL: 0.0787
  MSFT: 0.5240
  JPM: 0.0842
  XOM: 0.7279
  JNJ: -0.4148
Portfolio Daily Excess Return: 0.001350
Portfolio Daily Standard Deviation: 0.017506
Portfolio Annualized Sharpe Ratio: 1.2156

--- Minimum Variance Portfolio ---
Optimal Weights:
  AAPL: 0.2000
  MSFT: 0.2000
  JPM: 0.2000
  XOM: 0.2000
  JNJ: 0.2000
Portfolio Daily Excess Return: 0.000641
Portfolio Daily Standard Deviation: 0.010286
```

```
Expected Daily Excess Returns (mu):
 AAPL    0.000730
MSFT    0.000836
JPM     0.000552
XOM     0.001101
JNJ    -0.000016
dtype: float64

Covariance Matrix of Stock Excess Returns (sigma):
           AAPL      MSFT       JPM       XOM       JNJ
AAPL  0.000297  0.000205  0.000097  0.000057  0.000039
MSFT  0.000205  0.000299  0.000084  0.000030  0.000031
JPM   0.000097  0.000084  0.000227  0.000097  0.000042
XOM   0.000057  0.000030  0.000097  0.000315  0.000019
JNJ   0.000039  0.000031  0.000042  0.000019  0.000103
```

3. **Portfolio Factor Dependences (statsmodels.api):**

- Portfolio Excess Returns: Calculates the daily excess return of the *optimized portfolio* by taking a weighted average of the individual stock excess returns using the chosen_optimal_weights.
- FF3 Regression:
  - Prepares the independent variables (X_ff3) as the daily Mkt-Rf, SMB, and HML factors, adding a constant term (for alpha).
  - Runs an Ordinary Least Squares (OLS) regression of the portfolio's daily excess returns on these FF3 factors.
  - Prints the model_ff3.summary(), which includes the estimated betas (coefficients), their p-values, R-squared, and other statistical metrics.
- FF5 Regression:
  - Prepares X_ff5 with all five daily factors (Mkt-Rf, SMB, HML, RMW, CMA) and a constant.
  - Runs the OLS regression for FF5.
  - Prints model_ff5.summary().
- Tabular Comparison: Extracts the key betas and p-values from both FF3 and FF5 models and presents them in a pandas.DataFrame for easy comparison, including the alpha. It also shows the R-squared and Adjusted R-squared for both mode

```
Portfolio Daily Excess Returns Head:
 Date
2021-07-02    0.004534
2021-07-06   -0.018499
2021-07-07   -0.009164
2021-07-08   -0.009319
2021-07-09    0.016241
dtype: float64
```

```
--- FF3 Betas for the Optimal Portfolio ---
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.008
Model:                            OLS   Adj. R-squared:                  0.004
Method:                 Least Squares   F-statistic:                     1.996
Date:                Tue, 22 Jul 2025   Prob (F-statistic):              0.113
Time:                        22:55:10   Log-Likelihood:                 1978.5
No. Observations:                 752   AIC:                            -3949.
Df Residuals:                     748   BIC:                            -3930.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0014      0.001      2.188      0.029       0.000       0.003
Mkt-Rf        -0.0809      0.066     -1.229      0.219      -0.210       0.048
SMB           -0.2693      0.125     -2.146      0.032      -0.516      -0.023
HML            0.0151      0.128      0.118      0.906      -0.236       0.266
==============================================================================
```

```
--- FF5 Betas for the Optimal Portfolio ---
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.011
Model:                            OLS   Adj. R-squared:                  0.004
Method:                 Least Squares   F-statistic:                     1.629
Date:                Tue, 22 Jul 2025   Prob (F-statistic):              0.150
Time:                        22:55:10   Log-Likelihood:                 1979.5
No. Observations:                 752   AIC:                            -3947.
Df Residuals:                     746   BIC:                            -3919.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0015      0.001      2.300      0.022       0.000       0.003
Mkt-Rf        -0.0772      0.066     -1.172      0.242      -0.206       0.052
SMB           -0.2588      0.126     -2.057      0.040      -0.506      -0.012
HML            0.0020      0.128      0.015      0.988      -0.250       0.254
RMW           -0.1171      0.210     -0.558      0.577      -0.529       0.295
CMA           -0.2902      0.215     -1.352      0.177      -0.712       0.131
==============================================================================
Omnibus:                       16.714   Durbin-Watson:                   1.959
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               32.471
Skew:                          -0.032   Prob(JB):                     8.89e-08
Kurtosis:                       4.016   Cond. No.                         338.
```

```
--- Portfolio Betas Comparison (FF3 vs FF5) ---
        FF3 Beta FF3 p-value  FF5 Beta  FF5 p-value
Mkt-Rf -0.080915    0.219383 -0.077178     0.241724
SMB    -0.269267    0.032157 -0.258842     0.040052
HML     0.015141    0.905833  0.001968     0.987784
RMW          N/A         N/A -0.117135     0.577053
CMA          N/A         N/A -0.290214     0.176891
Alpha   0.001403    0.029002  0.001480     0.021698

--- FF3 Model Metrics ---
R-squared: 0.0079
Adjusted R-squared: 0.0040
MSE: 0.00030524

--- FF5 Model Metrics ---
R-squared: 0.0108
Adjusted R-squared: 0.0042
MSE: 0.00030518
```

Step 8: Interpretation of Results

Our analysis of the "Max Sharpe Ratio Portfolio composed of AAPL, MSFT, JPM, XOM, and JNJ" over the three-year period reveals that the Fama-French 5-factor model (FF5) generally provided a slightly improved explanation of its excess returns compared to the 3-factor model (FF3).

Looking at the model metrics, the adjusted R-squared for FF5 was 0.0108, which is slightly higher than 0.0042 from FF3. This indicates that the additional factors did meaningfully enhance the model's ability to explain the portfolio's return variability.

Specifically, when adding the Robust Minus Weak (RMW) and Conservative Minus Aggressive (CMA) factors in FF5, we observed that:

- The RMW factor had a negative and statistically significant beta of -0.117135 for the portfolio. This suggests our portfolio is negatively exposed to companies with robust profitability.
- The CMA factor had a negative and statistically significant beta of -0.290214. This implies the portfolio is negatively exposed to companies with aggressive investment policies.

The alpha (unexplained return) in the FF5 model was 0.001480, which was larger compared to the FF3 model's alpha of 0.001403. A smaller, less significant alpha suggests FF5 did a better job at explaining the systematic drivers of this portfolio's returns.

The shifts in the original FF3 betas (Mkt-Rf, SMB, HML) within the FF5 model were noticeable. This implies some level of correlation between the new factors and the original three, or that the new factors capture distinct dimensions of risk.

In conclusion, for this specific portfolio and time period, the FF5 model provided a better understanding of risk exposures and return drivers FF3. This highlights that while profitability and investment strategies are theoretically important, their empirical relevance can vary depending on the specific assets and market conditions.